



Hypothesis Testing for Poisson and Binomial Parameters · Worksheet 1

Further Statistics 1 Chapter 4: Hypothesis Testing for Poisson and Binomial Parameters. Answer all questions. Show your working clearly. Total: 66 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 4. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- In every test state H_0 and H_1 in terms of λ or p , show the probability you compare with the significance level, and write a conclusion in context. Give probabilities to 4 decimal places and actual significance levels as percentages.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Tests for the mean of a Poisson distribution	20	
B Critical regions and actual significance levels	21	
C Two-tailed tests and p-values	20	
Roar-level challenge	5	
Total	66	

Section A Tests for the mean of a Poisson distribution

20 marks

1. Emails used to arrive at an office at a mean rate of 5 per hour. The manager believes that the rate has increased. [2]

Write down suitable null and alternative hypotheses for a test of the manager's belief.

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2. The random variable $X \sim \text{Po}(\lambda)$. A single observation $x = 8$ is taken. [3]
Test, at the 5% level of significance, $H_0: \lambda = 4$ against $H_1: \lambda > 4$.

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3. The random variable $X \sim \text{Po}(\lambda)$. A single observation $x = 1$ is taken. [3]
Test, at the 5% level of significance, $H_0: \lambda = 6$ against $H_1: \lambda < 6$.

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4. Accidents at a junction used to occur at random at a mean rate of 1.5 per month. Traffic lights are installed, and in the following 3 months there is 1 accident. [4]
Test, at the 5% level of significance, whether the rate of accidents has decreased.

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5. A website receives complaints at random, at a mean rate of 3 per week.

- (a) State two conditions under which a Poisson distribution is a suitable model for the number of complaints received in a week. [2]

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- (b) After the website is redesigned, 6 complaints are received in a 4-week period. [4]
Test, at the 5% level of significance, whether the rate of complaints has decreased.

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- (c) Find the largest number of complaints in a 4-week period that would lead to the conclusion that the rate has decreased, at the 5% level. [2]

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(Total for Question 5 is 8 marks)

**Section B Critical regions and actual significance levels**

21 marks

6. The random variable $X \sim B(12, p)$. Find the critical region for a test of $H_0: p = 0.5$ against $H_1: p > 0.5$ at the 5% level of significance. [2]

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7. The random variable $X \sim \text{Po}(\lambda)$. Find the critical region for a test of $H_0: \lambda = 5$ against $H_1: \lambda > 5$ at the 5% level of significance, and state the actual significance level. [3]

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8. The random variable $X \sim B(20, p)$. Find the critical region for a test of $H_0: p = 0.3$ against $H_1: p < 0.3$ at the 5% level of significance, and state the actual significance level. [3]

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9. Cars pass a checkpoint at random, at a mean rate of 3 per hour. A researcher counts the cars passing in a 2-hour period to test whether the rate has increased, using a 5% level of significance. Find the critical region for the test and state its actual significance level. [4]

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10. In the past, 2% of the phone cases made by a factory have been faulty. After a new machine is installed, the quality manager believes that this proportion has increased. She takes a random sample of 150 phone cases.

(a) Write down suitable hypotheses, defining the parameter you use. [2]

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(b) Using a suitable approximation, find the critical region for a test of her belief at the 5% level of significance. [4]

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(c) State the actual significance level of the test. [1]

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(d) The sample contains 8 faulty phone cases. State, with a reason, the conclusion of the test in context. [2]

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(Total for Question 10 is 9 marks)

**Section C Two-tailed tests and p-values**

20 marks

11. Explain what is meant by the *p-value* of an observed test statistic, and state how it is used to decide whether to reject H_0 . [2]

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12. The random variable $X \sim \text{Po}(\lambda)$. A single observation $x = 9$ is taken. [3]
Test, at the 5% level of significance, $H_0: \lambda = 4$ against $H_1: \lambda \neq 4$.

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13. The random variable $X \sim \text{B}(20, p)$. A single observation $x = 2$ is taken. [3]
Test, at the 10% level of significance, $H_0: p = 0.25$ against $H_1: p \neq 0.25$.

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14. The random variable $X \sim \text{Po}(\lambda)$. A test of $H_0: \lambda = 8$ against $H_1: \lambda \neq 8$ is carried out at the 5% level of significance. [4]
Find the critical region, with the probability in each tail as close as possible to, but not exceeding, 2.5%. State the actual significance level.

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15. A spinner is designed so that it lands on red with probability 0.35. Priya believes that the probability is not 0.35. She spins the spinner 30 times and records the number of times, X , that it lands on red.

(a) Write down suitable hypotheses for Priya's test. [1]

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(b) Using a 5% level of significance, find the critical region for X , with the probability in each tail as close as possible to, but not exceeding, 2.5%. [4]

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(c) State the actual significance level of the test. [1]

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(d) The spinner lands on red 16 times. State, with a reason, the conclusion of the test in context. [2]

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(Total for Question 15 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. Faults occur on a production line at random, and it is claimed that they occur at a mean rate of 0.5 per hour. A manager wants to test $H_0: \lambda = 0.5$ against $H_1: \lambda < 0.5$, where λ is the mean number of faults per hour. She will observe the line for t hours, where t is an integer, and reject H_0 if no faults occur. [5]
- (a) Find the smallest value of t for which the significance level of the test is at most 5%, and state the actual significance level.
- (b) Instead, she observes for 12 hours and rejects H_0 if at most one fault occurs. Find the actual significance level of this test.

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END OF WORKSHEET

Total: 66 marks